

AGRICULTURAL SUPPLY CHAIN & POST-HARVEST LOSS MITIGATION

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FOCUS AREA: Sub-Saharan AgTech & Logistics Analytics

DOMAIN: Agricultural Supply Chain / GIS

PROBLEM STATEMENT

In Sub-Saharan AgTech and logistics analytics, the problem of high post-harvest perishability across smallholder horticultural supply chains leads to severe crop spoilage (30–50% produce loss), depressed farmgate earnings, and artificial market price volatility.

Currently, static distribution scheduling and fixed delivery routes are used, which fail because they operate without real-time integration of ambient thermal variables, corridor transport delays, and dynamic urban market price fluctuations.

We propose using geospatial network optimization, predictive regression modeling, and spatial-temporal feature engineering to predict shelf-life decay and dynamically route high-value perishable produce to optimal wholesale hubs, aiming to achieve a 15–25% reduction in total transit duration and a measurable reduction in overall post-harvest revenue loss.

KEY RESEARCH QUESTIONS

1. **Spoilage Predictability:** To what degree can historical transit time, road infrastructure rating, and ambient temperature/humidity metrics accurately forecast crop degradation rates for high-value perishables?
2. **Spatial Bottleneck Identification:** Where are the primary systemic delays along major agricultural transit corridors (e.g., farmgate-to-wholesale hubs)?
3. **Economic Impact:** What is the estimated revenue retention for smallholder farmer cooperatives if dynamic rerouting reduces total transit duration by 15–25%?

CORE PROJECT OBJECTIVES

- **Exploratory Data Analysis:** Map historical post-harvest loss percentages against transport corridor variables, storage types, and seasonal weather trends.
- **Predictive Spoilage Modeling:** Train regression and classification models to predict the shelf-life decay percentage of horticultural shipments as a function of transit exposure hours and climatic conditions.
- **Geospatial Corridor Optimization:** Develop a graph-based routing model that identifies optimal market destinations based on real-time commodity demand, pricing, and transit feasibility.
- **Decision Dashboard:** Design an interactive dashboard prototype to assist logistics managers in dispatching produce efficiently to minimize spoilage.

TECHNICAL APPROACH & METHODOLOGY

PHASE	ANALYTICAL FOCUS	METHODS & TOOLS	EXPECTED OUTPUTS
Data Ingestion	Data cleaning, spatial matching, timestamp alignment	Pandas, NumPy, GeoPandas	Cleaned master tabular & GIS dataset
Feature Eng.	Calculating thermal heat exposure during transit, route friction indices	Spatial Join, Haversine metrics, Feature Scaling	Engineered transit risk indicators
Modeling	Spoilage rate prediction & demand forecasting	Scikit-learn (Random Forest, XGBoost), Prophet	Evaluated ML models (RMSE, R², MAE)
Optimization	Shortest path & market price yield maximization	NetworkX, Linear Programming (PuLP)	Dynamic routing decision algorithm

DATASET REQUIREMENTS & SCHEMA OVERVIEW

The analysis will leverage a combination of open-source agricultural indicators, meteorological datasets, and open logistics telemetry:

- **FAOSTAT / Regional Ag-Data:** Historical post-harvest loss indices and regional production volumes.
- **Climate Data (CHIRPS / ERA5):** Ambient daily temperature, humidity, and rainfall indices along transport corridors.
- **Logistics Telemetry (Simulated/Open Source):** Transit duration, vehicle speed, stopover times, and transport vehicle types.

EXPECTED CAPSTONE DELIVERABLES

- **Comprehensive Data Pipeline:** Clean, documented Python codebase for ingestion, transformation, and model training.
- **Analytical Report:** Rigorous evaluation of post-harvest loss drivers with actionable policy and logistical recommendations.
- **Interactive Analytics Dashboard:** A Streamlit, Flutter, or Plotly Dash interface visualizing loss hotspots, live market price margins, and optimal dispatch routes.